## Week 5 Homework: Reading Planner

Implement the following four named functions in app.js. Each function must validate its inputs and return a result.

**1. plannedPages(days, dailyPages, restOnSeventhDay)**

Calculate the total number of pages that can be read over a given number of days.

- days: integer from 0 to 60.
- dailyPages: integer from 0 to 100.
- restOnSeventhDay: boolean.
- If enabled, every 7th day (7, 14, 21, ...) is a rest day with zero pages.
- Use a for loop and include the last day.
- Return null for invalid inputs. With valid inputs, zero days or zero daily pages returns 0.

**Examples:**

- plannedPages(7, 20, true) → 120
- plannedPages(7, 20, false) → 140
- plannedPages(61, 20, true) → null

**2. remainingPages(goalPages, readPages)**

Calculate how many pages remain to reach a reading goal.

- goalPages: integer from 1 to 5,000.
- readPages: integer from 0 to 6,000.
- Return 0 if the goal has been reached or exceeded.
- Return null for invalid inputs.

**Examples:**

- remainingPages(200, 79) → 121
- remainingPages(200, 220) → 0
- remainingPages(0, 0) → null

**3. daysToFinish(pagesLeft, dailyPages, restOnSeventhDay)**

Calculate the number of calendar days needed to finish the remaining pages.

- pagesLeft: integer from 0 to 5,000.
- dailyPages: integer from 1 to 100.
- restOnSeventhDay: boolean.
- Use a while loop until the total reading capacity reaches or exceeds pagesLeft.
- Count rest days when calculating calendar days.
- Return 0 when pagesLeft is 0 and the other inputs are valid.
- Return null for invalid inputs. There is no 60-day limit for this function.

**Examples:**

- daysToFinish(120, 20, true) → 6
- daysToFinish(121, 20, true) → 8
- daysToFinish(121, 20, false) → 7
- daysToFinish(121, 0, true) → null

**4. runReadingCommand(action, first, second, restOnSeventhDay)**

Create a function that calls the appropriate function based on action.

| action | Function | Parameters |
|---|---|---|
| "plan" | plannedPages() | first: days, second: daily pages |
| "remaining" | remainingPages() | first: goal pages, second: pages read |
| "finish" | daysToFinish() | first: pages left, second: daily pages |

- Accept only the exact strings "plan", "remaining", and "finish".
- restOnSeventhDay must be a boolean for every command, including "remaining".
- Return the result of the selected function.
- Return null for an unknown command or invalid input.
- Use if/else to select the appropriate function. Do not duplicate the calculations inside this function.

**Examples:**

- runReadingCommand("plan", 7, 20, true) → 120
- runReadingCommand("remaining", 200, 79, false) → 121
- runReadingCommand("finish", 121, 20, true) → 8
- runReadingCommand("Plan", 7, 20, true) → null

**Combining the Functions**

For a goal of 200 pages, 79 pages already read, a daily capacity of 20 pages, and rest days enabled:

1. Calculate pagesLeft using "remaining".
2. Pass the result to "finish" to calculate daysToFinish.
3. Separately calculate the pages readable in 7 days using "plan".

**Expected console output:**

pagesLeft: 121

daysToFinish: 8

plannedPagesIn7Days: 120

If pagesLeft is null, do not pass it to "finish". Print only "invalid-input" and stop the report. Also test this case with goalPages = 0.

General Requirements

- Use only the JavaScript concepts covered in class.
- Validate inputs before calculations and loops.
- Do not convert or automatically correct invalid inputs.
- Use return inside functions and print results only in the caller.
- Keep variables local to each function. Do not use shared state.
- Test valid inputs, invalid inputs, and boundary cases.
- Git is required.